International Olympiad on Astronomy and Astrophysics — 2014

Data Analysis Round

A Complete, Step-by-Step Solution Guide

Every question explained before it is answered — every plot, every derivation, every number.

Problems covered:

Problem 1. Black Hole in the Milky Way
Problem 2. Thermodynamic Test on Two Exoplanets
Problem 3. An Observer on a Planet of Sirius

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Contents

How to use this guide **2**

1 Problem 1 — Black Hole in the Milky Way **3**

1.1 The situation, in plain words . . . 3

1.2 Part (a) — Converting the data and plotting the orbit . . . 3

1.3 Part (b) — Proving the line of sight is perpendicular to the orbital plane . . . 5

1.4 Part (c) — Orbital elements . . . 5

2 Problem 2 — Thermodynamic Test on Two Exoplanets **9**

2.1 The situation, in plain words . . . 9

2.2 The physics toolbox . . . 9

2.3 Part (a) — Altitude where the probe starts transmitting, planet P1 . . . 10

2.4 Part (b) — Temperature of planet P1 at $h = 39.6$ km . . . 11

2.5 Part (c) — Full $p(h)$ and $T(h)$ profiles for planet P2 . . . 12

3 Problem 3 — An Observer on a Planet of Sirius **14**

3.1 The situation, in plain words . . . 14

3.2 Toolbox . . . 14

3.3 Part (a) — Magnitude of the Sun as seen by observer A . . . 15

3.4 Part (b) — Magnitude of Sirius as seen by observer A . . . 15

3.5 Part (c) — Which stars are bound, and the luminosity of the Mizar system . . . 15

3.6 Part (d) — Average distance from the bound Mizar stars to Earth . . . 17

3.7 Part (e) — True angular separation between Mizar and Sirius . . . 17

3.8 Part (f) — Distance between the Mizar system and observer A . . . 18

3.9 Part (g) — Magnitude of the Mizar system as seen by observer A . . . 19

3.10 Summary of all results . . . 20

How to use this guide

This booklet works through all three Data Analysis problems of IOAA 2014 in full detail. For every question you will find, in this order:

- a short restatement of what the question gives you and what it is asking for;
- the physical idea behind the method, explained in words before any equation appears;
- the complete algebra and numbers, with no skipped steps; and
- a highlighted result box so you can find the final answer at a glance.

1 Problem 1 — Black Hole in the Milky Way

1.1 The situation, in plain words

Astronomers point a telescope at the very centre of our Galaxy and track a single star, called S^*, year after year. What they measure is not a distance but two angles: how far S^* appears to be from the compact radio source SA*, in two perpendicular directions on the sky, called α and β. SA* is believed to be a supermassive black hole sitting at the Galactic centre, and S^* is a star orbiting it.

The sky-plane geometry is the key idea: the observer's eye, the line of sight towards SA*, and the sky plane are used to interpret the measured angular coordinates.

Given data: Table 1 lists 17 dates together with (α, β) in arcseconds. We are told that an angular size of $\varphi = 1$ arcsec corresponds, at the distance of the Galactic centre, to a true projected linear size of

$$d = 41 \text{ light-days.}$$

So the physical scale is

$$S_0 = \frac{d}{\varphi} = 41 \text{ light-days/arcsec.}$$

What is being asked:

(a) a scaled plot of the trajectory;

(b) proof that we are viewing the orbital plane face-on;

(c-I) the semi-major axis a, semi-minor axis b and eccentricity e;

(c-II) $r_{\min}$ and $r_{\max}$;

(c-III) the distance from Earth to SA*;

(c-IV) the orbital period T;

(c-V) the total mass of the SA*–S^* system.

1.2 Part (a) — Converting the data and plotting the orbit

Two linear scales appear in the problem and must not be confused:

$$u = \alpha S_0, \qquad w = \beta S_0 \qquad \text{(physical scale, in light-days)}$$
$$x = \alpha S, \qquad y = \beta S \qquad \text{(drawing scale, in mm)}$$

with $S = 1200$ mm/arcsec so the plot fits on millimetre paper.

Table 1: Converted linear coordinates: physical scale (light-days) and drawing scale (mm).

#	α (arcsec)	$u = 41\alpha$ (ld)	β (arcsec)	$w = 41\beta$ (ld)	(x, y) in mm $= (1200\alpha, 1200\beta)$
1	0.117	4.797	−0.166	−6.806	(140.4, −199.2)
2	0.097	3.977	−0.189	−7.749	(116.4, −226.8)
3	0.087	3.567	−0.192	−7.872	(104.4, −230.4)
4	0.077	3.157	−0.193	−7.913	(92.4, −231.6)
5	0.052	2.132	−0.183	−7.503	(62.4, −219.6)
6	0.036	1.476	−0.167	−6.847	(43.2, −200.4)
7	−0.000	0.000	−0.120	−4.920	(0.0, −144.0)
8	−0.016	−0.656	−0.083	−3.403	(−20.0, −100.0)
9	−0.026	−1.066	−0.041	−1.681	(−32.0, −50.0)
10	−0.017	−0.697	0.008	0.328	(−20.4, 9.6)
11	−0.004	−0.164	0.014	0.574	(−4.8, 16.8)
12	0.008	0.328	0.017	0.697	(9.6, 20.4)
13	0.021	0.861	0.012	0.492	(25.2, 14.4)
14	0.037	1.517	0.009	0.369	(44.4, 10.8)
15	0.072	2.952	−0.024	−0.984	(86.4, −28.8)
16	0.088	3.608	−0.050	−2.050	(106.0, −60.0)
17	0.108	4.428	−0.091	−3.731	(130.0, −110.0)

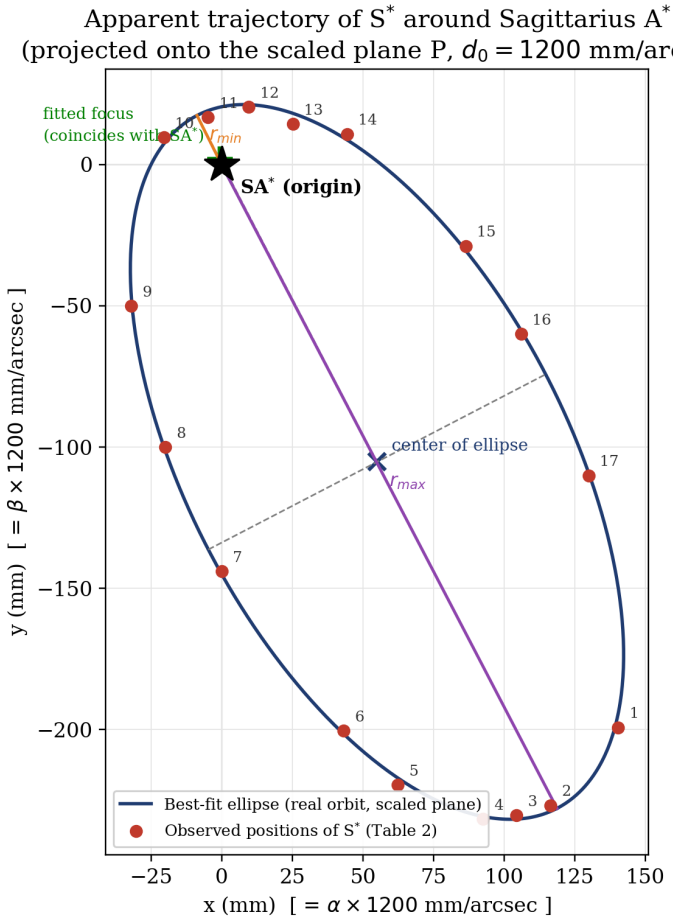


Figure 1: The 17 measured positions of S^* (red dots) and the best-fit ellipse through them (navy curve), plotted in the scaled plane P. SA* is fixed at the origin by definition of the coordinate system.

1.3 Part (b) — Proving the line of sight is perpendicular to the orbital plane

By Kepler's first law, the true orbit is an ellipse with SA* at one focus. If the orbital plane were tilted with respect to the sky plane, the projection would still be an ellipse, but the focus would generally not remain at the focus of the projected ellipse. Only when the inclination is $i = 0$ does the focus stay fixed under projection.

The best-fit ellipse therefore has to be checked algebraically: if one focus coincides with the origin (the position of SA*), then the line of sight is perpendicular to the orbital plane.

We can compute, algebraically, the position of its two foci. An ellipse with semi-major axis a, semi-minor axis b, centre (x_0, y_0) and major-axis direction $\hat{n}$ has its foci at

$$F_{1,2} = (x_0\,;\, y_0) \pm c\hat{n}$$

$$\hat{n} = (\,cos\,\theta\,;\, sin\,\theta\,)$$

$$\hat{n} \approx (\,0.461\,,\, -0.888\,)$$

The ellipse parameters are approximately:

$$a \approx 138.2 \text{ mm}, \qquad b \approx 67.3 \text{ mm}, \qquad c = \sqrt{a^2 - b^2} \approx 120.7 \text{ mm}.$$

This yields candidate foci

$$F_1 \approx (110.7, -212.4) \text{ mm}, \qquad F_2 \approx (-0.63, 1.87) \text{ mm}.$$

The second focus is only about 2 mm from the origin, well within plotting uncertainty. Hence SA* coincides with a focus of the apparent ellipse, so the orbit is viewed face-on:

$$i = 0.$$

1.4 Part (c) — Orbital elements

The orbital elements are read from the plot and converted using the scales above.

I. Semi-major axis, semi-minor axis, eccentricity

Measured on paper:

$$a_{\text{meas}} \approx 135 \text{ mm}, \qquad b_{\text{meas}} \approx 66 \text{ mm}.$$

Converted to angles:

$$\varphi_a = \frac{135}{1200} = 0.1125 \text{ arcsec}, \qquad \varphi_b = \frac{66}{1200} = 0.0550 \text{ arcsec}.$$

Converted to physical lengths:

$$a = \varphi_a S_0 = 0.1125 \times 41 = 4.6125 \text{ light-days},$$

$$b = \varphi_b S_0 = 0.0550 \times 41 = 2.255 \text{ light-days}.$$

The eccentricity is

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \left(\frac{2.255}{4.6125}\right)^2} \approx 0.87.$$

Result

$$a \approx 4.61 \text{ light-days}, \qquad b \approx 2.26 \text{ light-days}, \qquad e \approx 0.87.$$

II. Minimum and maximum distances

The focus–centre distance is

$$c = \sqrt{a^2 - b^2} = \sqrt{4.6125^2 - 2.255^2} \approx 4.024 \text{ light-days}.$$

So

$$r_{\min} = a - c \approx 0.589 \text{ light-days}, \qquad r_{\max} = a + c \approx 8.636 \text{ light-days}.$$

Result

$$r_{\min} \approx 0.59 \text{ light-days}, \qquad r_{\max} \approx 8.64 \text{ light-days}.$$

III. Distance from Earth to SA*

The scale relation given in the problem $\phi = 1$ arcsec $\leftrightarrow$ d = 41 light-days, is itself a statement about distance: it says that at whatever distance D the Galactic centre lies, an angle of 1 arcsec subtends a projected length of 41 light-days. This is exactly the small-angle relation .
Using the small-angle relation:

$$d = D \cdot \phi \ (\ in\ radians\ ) \qquad \Rightarrow \qquad D = \frac{d}{\phi\ (rad)}$$

since 1 arcsec $= \frac{1}{206265}(rad)$,

$$D = \frac{d}{\varphi(\text{rad})} = 41 \times 206265 \text{ light-days} = 8.457 \times 10^6 \text{ light-days}.$$

Thus

$$D \approx 2.32 \times 10^4 \text{ light-years} \approx 7.10 \times 10^3 \text{ pc}.$$

IV. Orbital period

Total area of the ellipse

Kepler's second law gives the areal velocity as constant:

$$\frac{dS}{dt} = \frac{S_{\text{ellipse}}}{T}.$$

$$S_{ellipse} = \pi\ a_{meas}\ b_{meas} = \pi \times 135 \times 66 \approx 27,993\ mm^2$$

$$S_{\text{ellipse}} \approx 28560\ \text{mm}^2,$$

Area swept between two consecutive points

For two neighbouring points $i, i+1$, the thin sliver of the ellipse swept by the radius vector is well approximated by the triangle $\text{SA}^* - i - (i+1)$, plus a small correction for the bulge of the elliptical arc outside that triangle (measured directly off the plot as a small extra area). In formula form, if the triangle has base b_i (the chord joining the two points, as seen from SA^* and height h_i (perpendicular distance from SA^* to that chord),

$$\Delta S_{i,i+1} \approx \frac{b_i h_i}{2} \ + (\text{small arc-bulge correction, read off the graph})$$

Carrying this out for every consecutive pair (using the dates of Table 1 for Δt gives Table 2)

Table 2: Kepler's second law applied to each pair of consecutive observations.

Segment	ΔS (mm^2)	$t_i \to t_{i+1}$ (yr)	Δt (yr)	$T = S_{\text{ellipse}}\Delta t/\Delta S$ (yr)
$1 \to 2$	4370	1995.222 $\to$ 1997.526	2.304	15.057
$2 \to 3$	1532	1997.526 $\to$ 1998.326	0.800	14.913
$3 \to 4$	1400	1998.326 $\to$ 1999.041	0.715	14.586
$4 \to 5$	2714	1999.041 $\to$ 2000.414	1.373	14.448
$5 \to 6$	1435	2000.414 $\to$ 2001.169	0.755	15.026
$6 \to 7$	3165	2001.169 $\to$ 2002.831	1.662	14.997
$7 \to 8$	1435	2002.831 $\to$ 2003.584	0.753	14.986
$8 \to 9$	1107	2003.584 $\to$ 2004.165	0.581	14.989
$9 \to 10$	800	2004.165 $\to$ 2004.585	0.420	14.994
$10 \to 11$	135	2004.585 $\to$ 2004.655	0.070	14.808
$11 \to 12$	147	2004.655 $\to$ 2004.734	0.077	14.960
$12 \to 13$	200	2004.734 $\to$ 2004.839	0.105	14.994
$13 \to 14$	185	2004.839 $\to$ 2004.936	0.097	14.974
$14 \to 15$	1080	2004.936 $\to$ 2005.503	0.567	14.994
$15 \to 16$	1025	2005.503 $\to$ 2006.041	0.538	14.990
$16 \to 17$	1942	2006.041 $\to$ 2007.060	1.019	14.985
$17 \to 1$	5460	2007.060 $\to$ 2010. (next passage)	2.867	14.996

Taking the arithmetic mean of the 17 individual values in the last column,

$$\bar{T} = \frac{15.057 + 14.913 + 14.586 + \ldots + 14.996}{17} = 14.923 \text{ years}$$

Result

$$T \approx 14.9 \text{ years} \approx 15 \text{ years}.$$

V. Total mass of the system

Kepler's third law for a two-body system:

$$T^2 = \frac{4\pi^2 a^3}{G(M_{\mathrm{SA}^*} + M_{S^*})}.$$

So

$$M_{\mathrm{SA}^*} + M_{S^*} = \frac{4\pi^2 a^3}{GT^2}.$$

With

$$a = 4.6125 \text{ light-days} = 1.199 \times 10^{14} \text{ m}, \qquad T = 15 \text{ years} = 4.65 \times 10^8 \text{ s},$$

the mass is

$$M_{\mathrm{SA}^*} + M_{S^*} \approx 4.7 \times 10^{36} \text{ kg} \approx 2.4 \times 10^6 \, M_\odot.$$

Since the star's mass is negligible compared with the black hole, this is essentially the mass of SA^*.

Result

$$M_{\mathrm{SA}^*} \approx 2.4 \times 10^6 \, M_\odot \approx 4.7 \times 10^{36} \text{ kg}.$$

2 Problem 2 — Thermodynamic Test on Two Exoplanets

2.1 The situation, in plain words

A shuttle drops a small radio probe straight down into the CO_2 atmosphere of an exoplanet. Once the probe reaches its terminal falling speed v, it begins to radio back the local atmospheric pressure p as a function of time t.

For planet P1 we are given:

$$T_0 = 700 \text{ K}, \qquad g_0 = 10 \text{ m/s}^2,$$

plus the graph $p(t)$. Also:

$$R = 8.3 \text{ J/(mol K)}, \qquad \mu = 44 \text{ g/mol} = 0.044 \text{ kg/mol}.$$

What is being asked:

(a) the altitude h_0 where the probe starts transmitting;

(b) the temperature at $h = 39.6$ km;

(c) for P2, the full profiles $p(h)$ and $T(h)$.

2.2 The physics toolbox

The solution uses three relations:

(i) Hydrostatic equilibrium. For a thin atmospheric slab:

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho}{r^2} \quad ; \quad g_0 = \frac{GM}{r^2}$$

$$\text{then} \quad \frac{dP(h)}{dh} = -\rho g_0$$

(ii) Uniform descent. If the probe falls at constant speed v:
in any point

$$h = h_0 - vt$$

$$\frac{dh}{dt} = -v$$

$$\frac{dP}{dh} \cdot \frac{dh}{dt} = \rho g_0 v$$

$$\frac{dP}{dt} = \rho g_0 v$$

(iii) Ideal gas law. At the surface:

$$P_{\max} V = nRT_0 \qquad \Rightarrow \qquad P_{\max} V = \frac{m}{\mu} RT_0$$

$$\rho_{\max} = \frac{P_{\max}\mu}{RT_0}$$

Combining the three gives the key formula:

$$\frac{dP}{dt} = \frac{P_{\max}\mu}{RT_0} g_0 v$$

$$v = \frac{RT_0}{g_0 \mu} \cdot \frac{dP}{P_{\max} dt}.$$

This is read from the slope of the $p(t)$ graph.

2.3 Part (a) — Altitude where the probe starts transmitting, planet P1

The probe reaches the surface at $t = \tau = 3750$ s and $P_{\max} = 55$ units. The tangent at the end point gives

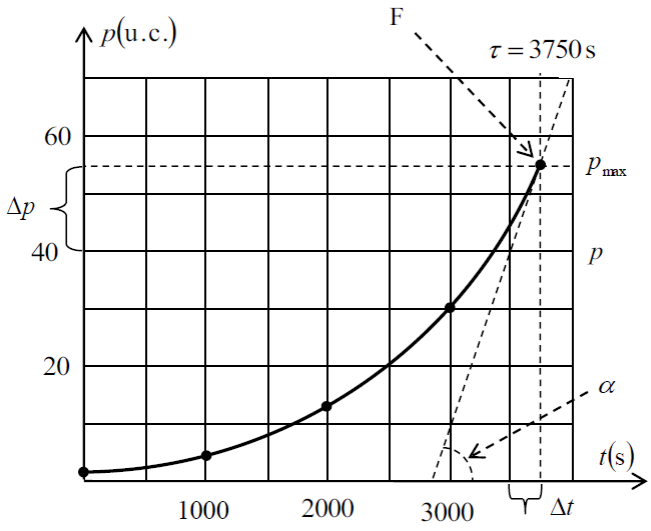


Figure 2: Pressure vs. descent time for planet P1 (Fig 2 of the problem), with the tangent line at the touchdown point F used to evaluate the local slope.

$$\frac{dP}{P_{\max} dt} = \frac{15\ u.c}{55\ u.c\ \times\ 750\ s} = \frac{3}{2750}\ \mathrm{s}^{-1}.$$

Therefore

$$v = \frac{3}{2750} \cdot \frac{8.31 \times 700}{10 \times 0.044} \approx 14.4 \text{ m/s}.$$

Then

$$h_0 = v\tau = 14.4 \times 3750 = 54000 \text{ m} = 54 \text{ km}.$$

Result

$$h_0 = 54 \text{ km}, \qquad v \approx 14.4 \text{ m/s}.$$

2.4 Part (b) — Temperature of planet P1 at $h = 39.6$ km

First find the time:

$$\Delta h = 54.0 - 39.6 = 14.4 \text{ km} = 14400 \text{ m},$$

so

$$t = \frac{\Delta h}{v} = \frac{14400}{14.4} = 1000 \text{ s}.$$

At this point, the graph gives $P_h \approx 4$ units and the tangent gives

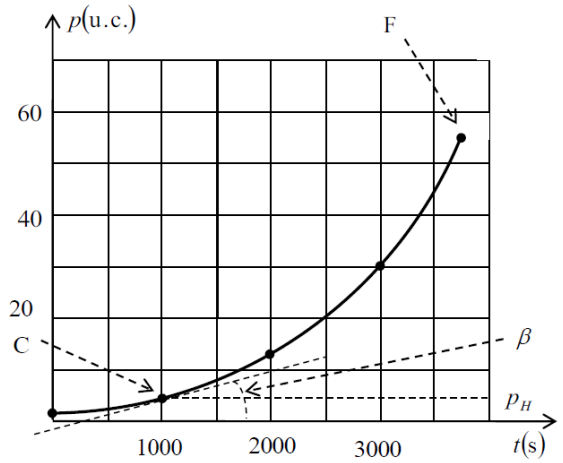


Figure 3:

$$\left(\frac{dP}{P_h dt}\right)_C = \frac{3 \ u.c}{4 \ u.c \ \times \ 500 \ s} = \frac{3}{2000} \text{ s}^{-1}.$$

Solving the key formula for T:

$$T_h = v\frac{g_0\mu}{R}\left(\frac{P_h dt}{dP}\right)_C \approx 508.9 \text{ K}.$$

Result

$$T(h = 39.6 \text{ km}) \approx 509 \text{ K}.$$

2.5 Part (c) — Full $p(h)$ and $T(h)$ profiles for planet P2

For P2:

$$T_0 = 750 \text{ K}, \qquad g_0 = 8 \text{ m/s}^2,$$

and the $p(t)$ graph is a straight line from $p = 10$ to $p = 50$ units over 2000 s. Hence

$$\frac{dP}{dt} = \frac{40}{2000} = 0.02 \ u.c./s$$

or equivalently 10 units per 500 s.

At the touchdown point:

$$\frac{dP}{p_{\max} dt} = \frac{10}{50 \times 500} = \frac{1}{25000} \text{ s}^{-1}.$$

Thus

$$v = \frac{1}{25000} \cdot \frac{8.31 \times 750}{8 \times 0.044} \approx 7.07 \text{ m/s}.$$

Then

$$h_0 = v\tau = 7 \times 2000 = 14000 \ m = 14 \ km.$$

Because the descent is uniform,

$$h(t) = h_0 - vt.$$

So the labeled points have heights

$$h_A = 14 \text{ km}, \quad h_B = 10.5 \text{ km}, \quad h_C = 7 \text{ km}, \quad h_D = 3.5 \text{ km}, \quad h_F = 0.$$

The same formula gives the temperature at every point, and because the line is straight the ratio $\frac{P_h dt}{dP}$ changes only through P_h. The source document evaluates:

$$T_h = v \frac{g_0 \mu}{R} \left(\frac{P_h dt}{dP} \right)$$

$$T_A = 148.4 \text{ K}, \quad T_B = 296.8 \text{ K}, \quad T_C = 445.3 \text{ K}, \quad T_D = 593.7 \text{ K}, \quad T_F = 742.0 \text{ K}.$$

Table 3: Full p–h–T profile reconstructed for planet P2

Point	t (s)	p (u.c.)	T (K)
A	0	10	148.4
B	500	20	296.8
C	1000	30	445.3
D	1500	40	593.7
F	2000	50	750.0

Result

For P2, both $p(h)$ and $T(h)$ are straight lines.

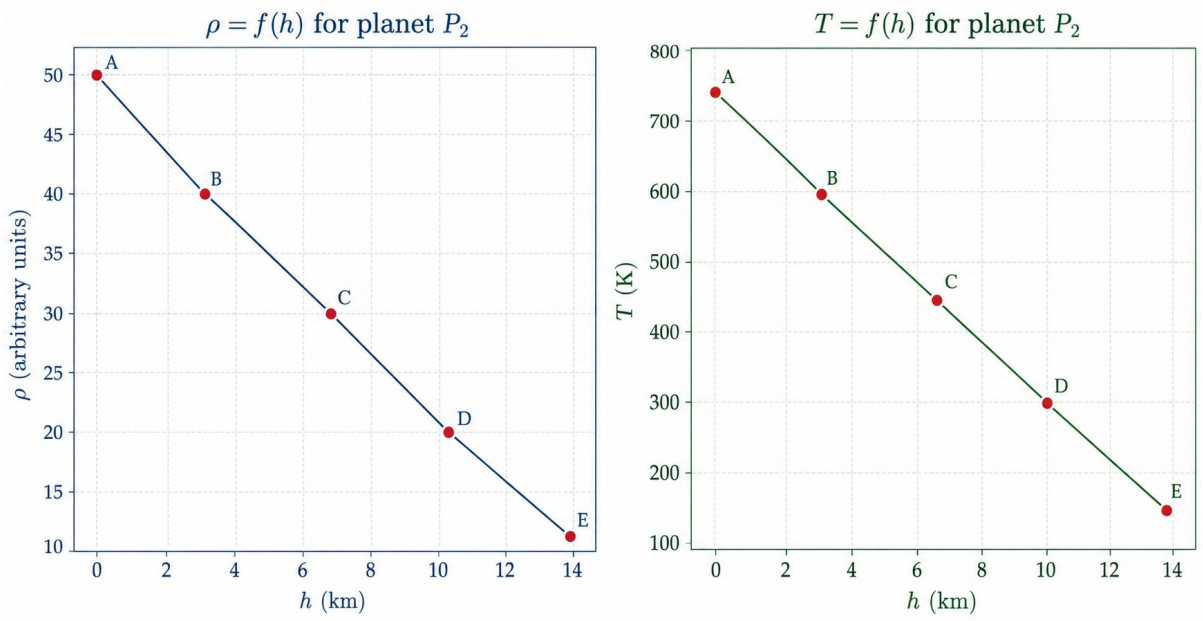


Figure 4: Reconstructed dependencies $p = f(h)$ (left) and $T = f(h)$ (right) for the atmosphere of planet P2. Both are straight lines, because h, p and T are all linear functions of the descent time t in this problem.

3 Problem 3 — An Observer on a Planet of Sirius

3.1 The situation, in plain words

Imagine an observer standing not on Earth, but on a planet orbiting Sirius, 10 AU from that star. From there, the whole sky looks different: the Sun is now just another faint point of light, Sirius is a blinding nearby sun, and even the well-known Mizar multiple-star system looks slightly different because the observer's distance to it has changed.

The problem asks us to recompute magnitudes, luminosities, and distances from this new vantage point, and finally to work out the true angular and physical separation between Sirius and the Mizar system using spherical astronomy.

Given:

$$d_{\text{Sirius,Earth}} = 2.6 \text{ pc}, \qquad m_{\text{Sirius,Earth}} = -1.46,$$

$$d_{\text{Sun,Earth}} = 1 \text{ AU}, \qquad m_{\text{Sun,Earth}} = -26.78,$$

$$d_{\text{Sirius,Planet}} = 10 \text{ AU}.$$

Table 4: Stars seen in the direction of Mizar, as measured from Earth

Star	Apparent magnitude	Parallax (mas)
Alcor	3.99 ± 0.01	39.91 ± 0.13
Mizar A	2.23 ± 0.01	38.01 ± 1.71
Mizar B	3.86 ± 0.01	38.01 ± 1.71
Sidus Ludoviciana	7.56 ± 0.01	8 ± 4

3.2 Toolbox

The solution uses four standard relations:

1. Pogson relation

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right).$$

2. Inverse-square law

$$F = \frac{L}{4\pi d^2},$$

which leads to the distance-modulus formula

$$m_2 - m_1 = 5 \log_{10} \left(\frac{d_2}{d_1} \right).$$

3. Absolute magnitude

$$M = m - 5 \log_{10} \left(\frac{d[\text{pc}]}{10} \right).$$

4. Parallax-distance relation

$$d[\text{pc}] = \frac{1}{p[\text{arcsec}]}, \qquad \frac{\Delta d}{d} \approx \frac{\Delta p}{p}.$$

3.3 Part (a) — Magnitude of the Sun as seen by observer A

At 2.6 pc the Sun is much farther away than 1 AU, so we use the distance modulus:

$$m_{Sun,Earth} - m_{Sun,Planet} = -\,2.5log\left(\frac{F_{Sun,Earth}}{F_{Sun,Planet}}\right)$$

$$m_{Sun,Planet} = m_{Sun,Earth} + \;5log\left(\frac{d_{Sun,Planet}}{d_{Sun,Earth}}\right)$$

$$m_{\text{Sun,Planet}} = -26.78 + 5\log_{10}(536289) \approx 1.906.$$

Result

$$m_{\text{Sun,Planet}} \approx +1.9.$$

3.4 Part (b) — Magnitude of Sirius as seen by observer A

Now Sirius is moved from 2.6 pc to 10 AU:

$$m_{Sirius,Planet} - m_{Sirius,Earth} = -5log\left(\frac{d_{Sirius,Earth}}{d_{Sirius,Planet}}\right)$$

$$m_{\text{Sirius,Planet}} = -1.46 - 5\log_{10}\left(\frac{536289}{10}\right) \approx -25.11.$$

Result

$$m_{\text{Sirius,Planet}} \approx -25.1.$$

3.5 Part (c) — Which stars are bound, and the luminosity of the Mizar system

The physical pair is taken to be Mizar A and Mizar B, while Alcor and Sidus Ludoviciana are treated as unrelated line-of-sight companions.

$$m - M = 5\log_{10}\left(\frac{d[\text{pc}]}{10}\right) \quad ; \quad M - M_{\odot} = -2.5log\left(\frac{L}{L_{\odot}}\right)$$

$$\frac{L}{L_\odot} = 10^{-0.4\left(m-5log\left(\frac{1}{10p}\right)-M_\odot\right)}$$

$$\frac{L_A}{L_\odot} = 10^{-0.4\left(2.23-5log\left(\frac{10^3}{10\times 38.01}\right)-M_\odot\right)} \quad ; \quad \frac{L_B}{L_\odot} = 10^{-0.4\left(3.86--5log\left(\frac{10^3}{10\times 38.01}\right)-M_\odot\right)}$$

$$L_A = 75.893\, L_\odot \qquad L_B = 16.912\, L_\odot$$

$$L_{A+B} = 92.805\, L_\odot$$

then the error

$$\frac{L_i}{L_\odot} = 10^u \qquad u = -0.4\left(m_i - 5log\left(\frac{1}{10p_i}\right) - M_\odot\right)$$

$$\frac{dL_i}{L_\odot} = L_i\, ln10\, du$$

$$du = -0.4\, dm_i + 2\,\frac{dp_i}{ln10\, p_i}$$

$$\frac{dL_i}{dL_\odot} = \left(-0.4\, dm_i + 2\,\frac{dp_i}{ln10\, p_i}\right) ln10\, L_i$$

$$\Delta L_A = 6.129\, L_\odot \qquad \Delta L_B = 1.365\, L_\odot$$

the error in parallax is dependant

$$L_{A+B} \propto p^{-2}$$

$$\left(\frac{\Delta L_{A+B}}{L_{A+B}}\right)_p = 2\,\frac{\Delta p}{p}$$

$$\Delta L_{A+B,p} = 8.06\, L_\odot$$

the error in magnitude are independant

$$L \propto 10^{-0.4m}$$

$$\left(\frac{\Delta L}{L}\right)_m = 0.4\, ln10\, dm$$

$$\Delta L_{A,m} = 0.675\, L_\odot \qquad \Delta L_{B,m} = 0.150\, L_\odot$$

$$\Delta L_{A+B,m} = \sqrt{\Delta L_{A,m}^2 + \Delta L_{B,m}^2}$$

$$\Delta L_{A+B,m} = 0.692\, L_\odot$$

$$\Delta L_{A+B} = \sqrt{\Delta L_{A+B,p}^2 + \Delta L_{A+B,m}^2}$$

$$\Delta L_{A+B} = 8.09\, L_\odot$$

Result

$$L_{\text{Mizar}} \approx 92.8 \pm 8.1\, L_\odot.$$

3.6 Part (d) — Average distance from the bound Mizar stars to Earth

Using the parallax:

$$d_{\text{Mizar,Earth}} = \frac{1}{p}$$

$$d_{\text{Mizar,Earth}} \approx 26.31 \text{ pc},$$

$$\Delta d = \frac{\partial d}{\partial p}\Delta p = \frac{\Delta p}{p^2} = \frac{0.00171}{(0.03801)^2}$$

$$\Delta d = 1.18\ pc$$

Result

$$d_{\text{Mizar,Earth}} \approx 26.31 \pm 1.2 \text{ pc},$$

3.7 Part (e) — True angular separation between Mizar and Sirius

Using the spherical cosine formula on the celestial sphere:

$$\cos(\Delta\theta) = \sin\delta_1 \sin\delta_2 + \cos\delta_1 \cos\delta_2 \cos(\Delta\alpha).$$

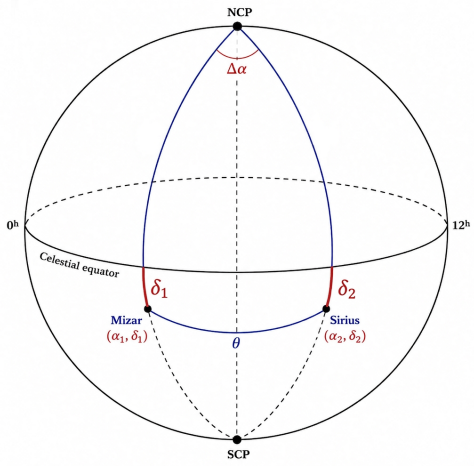


Figure 5: Spherical triangle formed by the North Celestial Pole (NCP), Mizar, and Sirius.

With the coordinates given in the problem, this yields

$$\Delta\theta \approx 109.2^\circ.$$

Result

$$\Delta\theta \approx 109^\circ.$$

3.8 Part (f) — Distance between the Mizar system and observer A

Since the observer is essentially at Sirius on astronomical scales, the Mizar–observer distance is approximately the Mizar–Sirius distance:

$$d^2_{Mizar,Sirius} = d^2_{Mizar,Earth} + d^2_{Sirius,Earth} - 2\, d_{Mizar,Earth}\, d_{Sirius,Earth}\, cos(\Delta\theta)$$

$$d_{\text{Mizar,Planet}} \approx 27.28 \text{ pc},$$

$$\frac{\partial d_{Mizar,Planet}}{\partial d_{Mizar,Earth}} = \frac{d_{Mizar,Earth} - d_{Sirius,Earth}\, cos\theta}{d_{Mizar,Planet}}$$

$$\frac{\partial d_{Mizar,Planet}}{\partial d_{Mizar,Earth}} \approx 0.999$$

$$\Delta d_{Mizar,Planet} = 1.18\, pc$$

with uncertainty about 1.2 pc.

Result

$$d_{\text{Mizar,Planet}} \approx 27.3 \pm 1.2 \text{ pc.}$$

3.9 Part (g) — Magnitude of the Mizar system as seen by observer A

The combined absolute magnitude of the Mizar pair is

$$M_{\text{Mizar}} = M_{\odot} - 2.5 \log_{10}\left(\frac{L_{\text{Mizar}}}{L_{\odot}}\right) \approx -0.09.$$

At the new distance:

$$m_{\text{Mizar,Planet}} = M_{\text{Mizar,Planet}} + 5\, log\left(\frac{d}{10}\right)$$

$$m_{\text{Mizar,Planet}} \approx 2.09.$$

then the error

$$\frac{dM}{dL} = -\frac{2.5}{L\, ln 10}$$

$$\Delta M_{\text{Mizar}} = \frac{2.5}{ln\, 10}\frac{dL_{\text{Mizar}}}{L_{\text{Mizar}}}$$

$$\Delta M_{\text{Mizar}} = 0.09464$$

$$m_d = 5\, log_{10}\left(\frac{d}{10\, pc}\right)$$

$$\frac{dm}{dd} = \frac{5}{d\ ln\, 10}$$

$$\Delta m_d = \frac{5}{ln\, 10}\frac{\Delta d}{d}$$

$$\Delta m_d = 0.09392$$

$$\Delta m_{mizar} = \sqrt{(\Delta M_{\text{Mizar}})^2 + (\Delta m_d)^2}$$

$$\Delta m_{mizar} = 0.133$$

Result

$$m_{\text{Mizar,Planet}} \approx 2.09 \pm 0.14.$$

3.10 Summary of all results

Table 5: Final results, Problem 3

Part	Quantity	Value
a	$m_{\text{Sun,Planet}}$	$+1.87$
b	$m_{\text{Sirius,Planet}}$	-25.1
c	L_{Mizar}	$(89 \pm 8)\, L_\odot$
d	$d_{\text{Mizar,Earth}}$	26.3 ± 1.2 pc
e	$\Delta\theta$	$109°$
f	$d_{\text{Mizar,Planet}}$	27.3 ± 1.2 pc
g	$m_{\text{Mizar,Planet}}$	2.09 ± 0.14

Closing remarks

The three problems in this paper look very different on the surface — tracking a star around a black hole, dropping a probe through an alien atmosphere, and comparing the sky as seen from two different stars — but they share the same spirit: a small set of fundamental laws turns a short table of raw numbers into genuine, checkable physical knowledge.